Rail and the c_0/K Limit

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Abstract

Rail does not beat the schedule problem; it dissolves it. The relevant number is not capacity but fixed cost per unit of capacity, c_0/K, and the load-factor threshold $\lambda^* = c_0/[K(p-c_v)]$ it implies. Rail drives c_0/K lower than any other mode, collapsing the break-even load factor to the point where the timetable is free. This note derives the threshold, tabulates the illustrative arithmetic across matatu, bus, passenger train, and block freight, explains why freight is the purest case, extends the threshold to the infrastructure layer with $\lambda^{**} = (c_0 + f_{\text{infra}}/N_{\text{departures}})/[K(p-c_v)]$, and shows that the failure mode—an empty train—is amplified, not escaped, at the extreme the consolidation constant so favours.

Yes — with one correction that makes the point sharper. It isn't capacity alone that saves the train. It is fixed cost per unit of capacity, c_0/K. Rail drives that ratio down further than any other mode, and the load-factor threshold collapses as a result.

1 The Number That Changes Everything

$$\lambda^* = \frac{c_0}{K\,(p - c_v)} \tag{1}$$

The bus needs a high load factor because c_0/K is large: one driver, one engine, one departure cost spread over 60 seats. The train spreads a similar departure cost over 1,000 seats—and the crew, the locomotive, and the track-access charge do not scale with the number of cars. Doubling the consist roughly doubles K while raising c_0 by maybe a third.

Illustrative arithmetic:

Mode	c_0 per departure	K	Margin/unit	λ^*
14-seater matatu	∼2,500	14	∼90	∼1.0 (needs full)
60-seat bus	∼5,000	60	∼90	∼0.93
Passenger train (trunk)	∼400,000	1,300	∼400	∼0.33
Block freight train	∼600,000	3,000 t	∼1,500	∼0.10

The bus cannot run a schedule, because its threshold is ∼93% and any off-peak run loses money. The train can run a schedule, because its threshold is 10–33% and a half-empty train still clears it.

This is why the schedule is viable for rail and pathological for buses. Not better management—different arithmetic.

2 The Three-Way Comparison

The matatu waits because it must. The bus is forbidden to wait and must lose money. The train does not need to wait—the threshold is already so low that the schedule is free.

That is the deepest point: the train escapes the bus's trap not by being allowed to consolidate, but by not needing to.

	Matatu	Bus	Train
λ^*	$\sim$1.0	$\sim$0.93	0.10–0.33
Waiting allowed?	Yes	No	Irrelevant
Behaviour	Waits to fill	Runs empty	Runs to schedule
Outcome	Spot market, low fare	Subsidy	Profitable where density holds

3 Why Freight Is the Purest Case

Freight has a higher margin per unit and a higher capacity per departure. Coal, ore, and grain trains are the limit case: one origin, one destination, no intermediate stops, no LTL, no passenger service obligations. λ^* is effectively near zero.

This is why freight rail is profitable almost everywhere it exists, and passenger rail is not. Same track, same c_0 structure—different margin per unit. Indian Railways, JR Freight, and the US Class I railroads all demonstrate the same thing: the freight side clears the threshold trivially, the passenger side does not, and the freight side cross-subsidises the passenger side or the state does.

4 The Infrastructure Layer

The formula above is the operating break-even. The full economic break-even adds track:

$$\lambda^{**} = \frac{c_0 + f_{\text{infra}}/N_{\text{departures}}}{K\,(p - c_v)} \tag{2}$$

This is why frequency matters more for rail than for any other mode. A line that cost billions amortises its infrastructure over every departure it ever runs. A four-trains-a-day line pays the same capital cost as a forty-trains-a-day line, spread over a tenth of the volume. The high-frequency trunk clears λ^{**}; the low-frequency rural line never does, no matter how full each train is.

5 The Failure Mode Is Also Amplified

High c_0 cuts both ways. A train that runs empty is a catastrophic loss, not a small one. The kill rule matters more for rail than for buses—and the coverage mandate forbids it more strongly. Amtrak's long-distance trains are the canonical example: they run because the mandate requires it, and the loss per passenger-mile is enormous precisely because K is so large and the train so empty.

Rail does not escape the consolidation constant. It sits at the extreme where the constant is most favourable—and where violating it is most expensive.

6 The One Line

Trains do not beat the schedule problem; they dissolve it. The enormous capacity drives c_0/K down until the break-even load factor is so low that the schedule costs nothing. The matatu waits because it must, the bus loses money because it cannot, and the train runs to timetable because—for the first time in the hierarchy—the timetable is free.